

plasma can be shaped into specific geometry to match the needs of intended application. This is done by applying magnetic field which will make a cyclic motion of the charged particle. This is also an important property as it frees the traditional image of antennae in our mind, although geometries are limited.

The plasma antenna

Antenna is a matching section between two output terminals of a transmitter and space, or between space and two input terminals of a receiver. Radiation is simply the transfer of energy through a medium. So a plasma antenna is a type of transmission and reception device that makes use of plasma medium rather than metal components.

The electron behaviour of plasma antenna is completely different from that of metal antenna. In the plasma antenna, the functioning concept is altogether different. It is due to 'electrons in free space' rather than 'electrons moving freely.' The design allows for extremely short pulses that are very important to many forms of digital communication and radars.

One fundamental distinguishing feature of a plasma antenna is that the gas ionising process can manipulate resistance. When de-ionised, it has infinite resistance and hence it does not react with RF. When ionised, it will have some resistance due to which it will react with EM waves.

Methods of producing radiations

There are two methods of producing radiation:

1. m-radiation method. In this method, the radiation is produced by current oscillations on the surface of a metal and by disturbing current on the interface between plasma and medium. (Example is surface wave-driven plasma antenna.)

2. d-radiation method. In this method, the excitation is applied to the interface which disturbs current between plasma and medium and

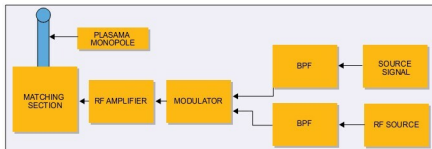


Fig. 4: Block diagram of simple monopole plasma antenna

radiation takes place. It is just like the radiation in dielectric antennae.

Plasma antennae are of two types:

1. Gas chamber. In this type, we use the DC discharge system with a very high-voltage source applied to cathode and anode, and then the signal is superimposed on it (a plasma column). This is a primitive implementation of plasma antenna.

2. Solid-state semiconductor type. The concept of this type of plasma antenna is that the charge carriers in metal and semiconductor behave similar to those in gas plasma. The medium properties will vary as per constructions. However, the interaction of EM waves with charge carriers will have very similar properties to that of quasi-neutral particles.

The semiconductor having enough free carriers to interact with EM waves is called semiconductor plasma with very high electron density that can be obtained by heating, current injection or by optical excitation.

Features and applications

Salient features and applications of some of the plasma antennae are:

1. Reflectors. As stated earlier, if EM wave frequency is smaller than plasma frequency, the wave will be reflected or absorbed. This feature has been exploited in the design of radar-absorbing materials for stealth applications.

2. Windowing. It is a term coined for RF signals being transmitted through plasma tubes which are off or low enough in plasma density. This feature has been exploited in side-lobe reduction and broadband jamming equipment.

3. Stacking. Plasma antennae can be stacked into one another for different frequencies. For instance, the inner antenna is optimised for high frequency and outer antenna is optimised for low frequency, so both of them will work independent of each other. Also, the plasma antennae are more susceptible to frequencies not detected by metal ones.

4. In war. Plasma antennae do not melt, have heat and fire resistance and, as ohmic losses are very less, they have wider range of power-handling capability. They can be easily prepared for use in wars.

5. Car anti-collision radar system. Perhaps this is the most awaited commercial application of any antenna system. Plasma antennae solve this problem. The optically-excited semiconductor antenna array combined with adequate logic control in California Institute of Technology resulted in such a system. However, this is still in its early phase and prototypes have been developed. Just think of its application in low-visibility areas!

6. Mobile industry. If the plasma silicon antennae are used in towers, the towers would be able to transmit denser beams and provide more phone support than traditional antennae, though this could make humans and animals more prone to radiation-related problems. But due to eradication of network interference, this would directly reduce the spectrum cost. HF CDMA plasma antennae can have low probability of interception, which is an important parameter of CDMA communication. ●

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